

Color image Processing & Compression

Q.1) image resolution = 640×480 pixels

bits per color channel = 8 bits

RGB pixel = (R=120, G=200, B=80)

Compressed image size = 450 KB

Frame rate = 25 frames/sec

Find

a) RGB $\rightarrow$ Gray conversion

Standard luminance equation

$$\text{Grayscale} = 0.299R + 0.587G + 0.114B$$

$$= 0.299(120) + 0.587(200) + 0.114(80)$$

$$= 35.88 + 117.4 + 9.12$$

$$231.4 \approx 162.5 \approx 162 \times 0.001 = 219 \times 10^{-3}$$

b) RGB $\rightarrow$ HSV conversion

Normalize RGB values

$$R = \frac{120}{255} = 0.47, \quad G = \frac{200}{255} = 0.78, \quad B = \frac{80}{255} = 0.31$$

$$\text{Max} = 0.78, \quad \text{Min} = 0.31$$

$$\Delta = \text{max} - \text{min} = 0.47$$

$$\text{saturation} = \frac{\Delta}{\text{max}} = \frac{0.47}{0.78} = 0.60$$

Hue (H):

max = G

$$H = 60^\circ \times \left[2 + \frac{(B-R)}{0.47} \right]$$

$$= 60 \times \left[2 + \frac{0.31 - 0.47}{0.47} \right]$$

$$= 60 \times (2 - 0.34)$$

$$= 99.6^\circ$$

c) image size before compression

$$\text{total pixels} = 640 \times 480 = 307,200 \text{ samples}$$

$$\text{Bit Per Pixel} = 8 \times 3 = 24 \text{ bits}$$

$$\text{Total bits} = 307,200 \times 24 = 7,372,800 \text{ bits}$$

$$\text{convert into MB: Bytes} = \frac{7,372,800}{8} = 921,600$$

$$\text{MB} = \frac{921,600}{1024^2} = 0.88 \text{ MB}$$

d) compression Ratio

$$= \frac{\text{original size}}{\text{compressed size}}$$

convert original size to KB $= 0.88 \times 1024$
 $= 901 \text{ KB}$

$$CR = \frac{901}{450} \approx 2.0022$$

e) Bit Rate Calculation

convert compressed size to bits 450 KB

$$450 \times 1024 \times 8 = 3686400 \text{ bits}$$

$$\text{Bit rate} = 3686400 \times 25$$

$$= 92160000 \text{ bit/sec}$$

$$= 892.16 \text{ mbps}$$

Q.2) Spatial Resolution = 1024×768 Pixels
 Bits Per Pixels = 6 bits
 max signal Fz. = 4 kHz
 sampling Fz. = 7 kHz

total number of pixels = $1024 \times 768 = 786432$ pixels

Gray level = $2^6 = 2^6 = 64$

Intensity range: 0 to (64-1)

: 0 to 63

64 levels

c) memory required to store image

$$\text{Total bits} = 786432 \times 6$$

$$= 4718592 \text{ bits}$$

convert bit to byte

$$= \frac{4718592}{8}$$

$$= 589824 \text{ bytes}$$

$$\text{Convert KB} = \frac{589824}{1024} = 576 \text{ KB}$$

$$\text{Convert MB} = \frac{576}{1024} = 0.5625 \text{ MB}$$

d) aliasing check using sampling theorem.

$$f_s \geq 2f_{\max}$$

$$2f_{\max} = 2 \times 4 = 8 \text{ KHz}$$

$$7 \text{ KHz} \leq 8 \text{ KHz}$$

e) Pixel connectivity = 4 = 4 = 4 = 4

- 4 connected (Pixel)

$$p - \begin{matrix} p \\ | \\ p \end{matrix} - p$$

- 8 connected Pixels

$$\begin{matrix} p & p & p \\ p & x & p \\ p & p & p \end{matrix}$$

$$(x-m) \times (n-l) \times \dots = \dots$$

$$d = s_0 + s_1 + s_2 = 102$$

$$u = (1)(5) + (5)(1) = 102$$

$$l = (1)(1) = 102$$

Q.3) $x(n) = \{1, 2, 1\}$
 $h(n) = \{2, 1, -1\}$

$$x(n), h(n), d(n), u(n), l(n) = \dots$$

a) Linear convolution

length $L = 3 + 3 - 1 = 5$

$$\begin{matrix} \begin{matrix} 001 & 091 & 001 \\ 021 & 081 & 011 \\ 1001 & 001 & 051 \end{matrix} \end{matrix} = \dots$$

$$y(0) = 1 \cdot 2 = 2$$

$$y(1) = 1 \cdot 1 + 2 \cdot 2 = 5$$

$$y(2) = 1 \cdot (-1) + 2 \cdot 1 + 1 \cdot 2$$

$$= -1 + 2 + 2 = 3$$

$$y(3) = 2 \cdot (-1) + 1 \cdot 1 = -2 + 1 = -1$$

$$y(4) = 1 \cdot (-1) = -1$$

$$\frac{0511}{p}$$

$$y(n) = \{2, 5, 3, -1, -1\}$$

b) 3 point circular convolution

$$y(n) = \sum_{k=0}^{2} x(k) h(n-k) \text{ mod } 3$$

$$\dots$$

$$y(0) = 1(2) + 2(-1) + 1(1) = 1$$

$$y(1) = 1(1) + 2(2) + 1(-1) = 4$$

$$y(2) = 1(-1) + 2(1) + 1(2) = 3$$

$$y(n) = \{1, 4, 3\}$$

c) auto correlation of $x(n)$

$$R_{xx} = \sum x(n) x(n-k)$$

$$R(0) = 1^2 + 2^2 + 3^2 = 14$$

$$R(1) = (1)(2) + (2)(1) = 4$$

$$R(2) = (1)(1) = 1$$

$$R_{xx} = \{14, 4, 1\}$$

d) mean filter operation

$$\text{image} = \begin{bmatrix} 100 & 120 & 140 \\ 110 & 130 & 150 \\ 120 & 140 & 160 \end{bmatrix}$$

$$\text{mean filter} = \frac{\text{sum of pixel}}{3} = \frac{9+5+1}{3}$$

$$= \frac{15}{3}$$

$$= 5 = \text{center pixel value}$$

e) Histogram Equalization

$$\text{Gray level} = 64, (x-m) \text{ and } (x) \text{ etc. } \sum_{i=0}^{63} 1 = 64$$

$$\text{total pixels} = 4096$$

$$\text{pixels per gray level} = \frac{4096}{64} = 64$$

$$P = \frac{1}{64} + \frac{1}{64} + \dots + \frac{1}{64} = \frac{1}{64}$$

$$E = \frac{1}{64} + \frac{1}{64} + \dots + \frac{1}{64} = \frac{1}{64}$$

$$E = \frac{1}{64} + \frac{1}{64} + \dots + \frac{1}{64} = \frac{1}{64}$$

$$CDF = (20 + 1) \times 64 = 1344$$

CDF up to gray level 20: 1344

$$CDF = (20 + 1) \times 64 = 1344$$

$$S_k = \frac{CDF}{\text{Total Pixel}} \times (L-1)$$

$$= \frac{1344}{4096} \times 63 \approx 20$$

new gray level: 20

Q.4) Passband edge frequency: $f_p = 1 \text{ kHz}$

Stopband edge frequency: $f_s = 1.5 \text{ kHz}$

sampling $f_s = f_{sa} = 8 \text{ kHz}$

min stopband attenuation = 40 dB

a) normalized digital passband and stopband f_{n0}

$$\omega = \frac{2\pi f}{f_{sam}}$$

$$\text{Passband } f_{pe} = \omega_p = \frac{2\pi \times 1000}{8000} = \frac{\pi}{4} \text{ rad/sample}$$

$$\text{Stopband } f_{se} = \omega_s = \frac{2\pi \times 1500}{8000} = \frac{3\pi}{8} \text{ rad/sample}$$

$$\left. \begin{aligned} \sin^2 \frac{\omega}{2} &= \frac{(1 - \cos \omega)}{2} \\ \cos \omega &= 1 - 2 \sin^2 \frac{\omega}{2} \end{aligned} \right\} \text{ (1)}$$

b) selection of window function

- required stopband attenuation 40 dB

Hamming window 41 dB attenuation

c) Transition width and filter order

Transition width: $\Delta\omega = \omega_s - \omega_p$

$$= \frac{3\pi}{8} \frac{\pi}{4}$$

$$\Delta\omega = \pi/8 \text{ rad/s}$$

- FIR filter order (Hamming window)

$$N \approx \frac{3.3\pi}{\Delta\omega} = \frac{3.3\pi}{\pi/8} \approx 26.4$$

Filter length $= N+1 = 28$

d) Ideal impulse response of low pass filter

cut off freq.

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{\pi/4 + 3\pi/8}{2} = \frac{5\pi}{16}$$

ideal impulse response

$$h_d(n) = \begin{cases} \frac{\sin(n\omega_c/2)}{n\omega_c/2}, & n \neq N/2 \\ \omega_c/2, & n = N/2 \end{cases}$$

e) effect of window on ideal impulse response.

- infinite length

f) Advantages of FIR over IIR Filter.

- exact linear phase

Q.5) Binary image $A = \begin{bmatrix} 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Structure element $B(3 \times 3)$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

a) Dilation $(A \oplus B)$

Rule: sets the output pixel to 1 if any pixel under B is 1.

$$A \oplus B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

b) Erosion $A \ominus B$

Rule: Erosion sets the output pixel to 1 only if all pixels under B are 1.

$$A \ominus B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

c) Opening $A \circ B$

$$A \circ B = (A \ominus B) \oplus B \quad \because A \ominus B = 0$$

$$A \circ B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \oplus B = B$$

d) Closing $A \bullet B$

$$A \bullet B = (A \oplus B) \ominus B$$

$$A \bullet B = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \ominus B = A$$

e) Boundary Extraction

- finds the outlines of objects in a binary image.

$$\text{Boundary}(A) = A - (A \ominus B)$$

Q.6]

$$x(n) = \{1, 0, 2, 0\} \quad n = 0, 1, 2, 3$$

Number of points $(N = 4)$

$$\text{Let } \omega_4 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = -j$$

a) 4 Point DFT of $x(n)$

DFT Formula: $X(k) = \sum_{n=0}^{N-1} x(n) \omega_N^{kn}$

$$X(k) = \sum_{n=0}^3 x(n) \omega_4^{kn}$$

$$k=0 \quad x(0) = 1+0+2+0 = 3$$

$$k=1 \quad x(1) = 1 + 2\omega_4^2 = e^{-j\pi} = -1$$

$$= 1 + 2(-1) = -1$$

$$k=2 \quad x(2) = 1 + 2\omega_4^4 = 1 + 2 = 3$$

$$k=3 \quad x(3) = 1 + 2\omega_4^6$$

$$= 1 - 2 = -1$$

$$X(k) = \{3, -1, 3, -1\}$$

b) 4 point IDFT and verification

IDFT Formula

$$x(n) = \frac{1}{4} \sum_{k=0}^3 x(k) W_4^{-kn}$$

$$x(k) = \{3, -1, 3, -1\}$$

$$x(0) = \frac{1}{4} (3 + 1 + 3 + 1) = 2$$

$$x(1) = \frac{1}{4} (3 + j - 3 - j) = 0$$

$$x(2) = \frac{1}{4} (3 + 1 + 3 + 1) = 2$$

$$x(3) = \frac{1}{4} (3 - j - 3 + j) = 0$$

$$x(n) = \{2, 0, 2, 0\}$$

c) 4 point FFT using DIT

$$E = 0 + 2 + 0 + 1 = (0)x \quad 0 = 1$$

- separate even and odd samples

$$F = 1 + 3 + 1 = (1)x \quad 1 = 3$$

$$x_e = \{1, 2\} \quad x_o = \{0, 3\}$$

- 2 point DFTs

$$G = 1 + 3 = (3)x \quad 3 = 1$$

$$F = 3 + 1 =$$

$$x_e = \{1 + 2, 1 - 2\} = \{3, -1\}$$

$$H = 1 + 3 = (3)x \quad 3 = 1$$

$$x_o = \{0 + 3, 0 - 3\} = \{3, -3\}$$

$$I = 3 - 1 =$$

$$J = 1 - 3 = (1)x$$

- Combine

$$x(k) = x_e(k) + w_k^* x_o(k)$$

$$x(0) = 3, x(1) = -1$$

$$x(2) = 3$$

$$x(3) = -1$$

$$\text{FFT output} = \{3, -1, 3, -1\}$$

d) verification of DFT linearity

$$x_1(n) = \{1, 0, 1, 0\}$$

$$x_2(n) = \{0, 1, 0, 1\}$$

DFTs:

$$\text{DFT}[x_1] = \{2, 0, 2, 0\}$$

$$\text{DFT}[x_2] = \{2, 0, -2, 0\}$$

$$\text{DFT}[x_1] + \text{DFT}[x_2] = \{4, 0, 0, 0\}$$

$$\text{sum signal: } x_1 + x_2 = \{1, 1, 1, 1\}$$

$$\text{DFT}[x_1 + x_2] = \{4, 0, 0, 0\}$$